

**PROJECT REPORT OF INDUSTRY ORIENTED HANDS ON
EXPERIENCE (IOHE)**

ON

**MANNLYTICS : AI-Based Mental Health
Detection Using NLP On User-Generated Text**

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

We hereby certify that the work presented in the project report entitled “AI-Based Mental Health Detection Using NLP on User-Generated Text”, submitted in partial fulfilment of the requirements for the award of the degree of Bachelor of Engineering in Computer Science and Engineering, in the Department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, is an authentic record of our own work carried out under the supervision of Dr. Monika Aggarwal.

We further declare that the work embodied in this project report has been completed by us independently and has not been submitted, either in part or in full, to any other university or institute for the award of any degree, diploma, or other academic qualification.

The assistance and guidance received during the course of this project have been duly acknowledged. We have sincerely put in our efforts, knowledge, and dedication to complete this project successfully to the best of our abilities.

Place: Rajpura

Yashika Khanna ,Yashika Taneja, Upinder Kaur

Date: 15/05/2026

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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We would also like to acknowledge the contribution of various online resources, research materials, and open-source platforms that assisted us during the implementation and research phase of this project. In particular, the Kaggle dataset and different open-source tools proved highly beneficial in developing and testing our system.

Furthermore, we extend our heartfelt thanks to our family members and friends for their constant motivation, patience, encouragement, and moral support throughout the course of this project. Their support helped us remain focused and motivated during every stage of the project development.

Lastly, we are grateful to everyone who directly or indirectly contributed to the successful completion of this project.

ABSTRACT

The increasing prevalence of mental health disorders such as depression, anxiety, stress, bipolar disorder, personality disorders, and suicidal tendencies has created a strong need for early detection and timely intervention. In recent years, social media platforms and online communication channels have generated massive amounts of textual data that reflect users' emotions, thoughts, and psychological conditions. This textual information can be effectively analyzed using Natural Language Processing (NLP) techniques to identify patterns associated with mental health conditions.

This paper presents the implementation of a mental health detection model using a hybrid NLP-based approach for automated text analysis and classification. The proposed hybrid model combines VADER sentiment analysis, TF-IDF feature extraction, and Logistic Regression to classify users' mental states into multiple categories including anxiety, bipolar disorder, depression, stress, personality disorder, suicidal tendencies, and normal behavior using a multi-class classification technique. To improve contextual understanding and semantic interpretation of text, advanced Transformer-based architectures such as DistilBERT and RoBERTa are also implemented and evaluated.

The system was trained and tested on a dataset containing 52,678 labeled text instances collected from mental health-related discussions and social media content. Experimental results demonstrated that the proposed model achieved an accuracy of 89.97%, showing strong performance in identifying different psychological conditions from textual inputs. The combination of traditional machine learning techniques with deep learning-based Transformer models enhanced both feature representation and contextual understanding, leading to improved classification accuracy and reliability.

The proposed system offers a scalable, efficient, and high-quality solution for real-time mental health assessment based on textual information. It can assist healthcare professionals, counselors, and digital mental health platforms in the early identification of psychological disorders, enabling faster support, monitoring, and intervention for individuals at risk.

1. INTRODUCTION

Mental health disorders such as depression, anxiety, stress, and suicidal tendencies have seen a significant rise in recent years, becoming a major global concern. These conditions not only affect an individual's emotional and psychological well-being but also impact their social interactions, productivity, and overall quality of life. Early identification and timely intervention are crucial to prevent the worsening of such conditions. However, traditional methods of mental health assessment, such as clinical interviews and self-reporting, are often time-consuming, subjective, and not easily scalable to large populations. With the rapid growth of social media platforms and digital communication, a massive volume of textual data is generated daily. People frequently express their thoughts, emotions, and personal experiences through posts, comments, and messages. This data provides a valuable opportunity to analyze human behavior and detect psychological patterns using computational techniques. Leveraging such data can enable automated, real-time mental health monitoring systems that complement traditional diagnostic approaches.

This project focuses on the development of a mental health detection system using a hybrid Natural Language Processing (NLP) approach. The proposed system analyzes user-generated text to identify potential mental health conditions. It combines multiple techniques, including VADER for sentiment analysis, TF-IDF for feature extraction, and Logistic Regression for classification. These methods allow the system to capture both emotional tone and textual patterns effectively.

In addition to traditional machine learning techniques, advanced Transformer-based models such as DistilBERT and RoBERTa are incorporated to improve contextual understanding and semantic interpretation of text. These models enhance the system's ability to detect subtle linguistic cues and complex emotional expressions that are often missed by conventional approaches.

The system is trained on a dataset of 52,678 labeled instances covering multiple mental health categories, including anxiety, bipolar disorder, depression, stress, personality disorder, suicidal tendencies, and normal mental state. The proposed model achieves an overall accuracy of 89.97%, demonstrating its effectiveness in classifying mental health conditions based on textual input.

Furthermore, the system is designed to provide additional insights such as risk scores, confidence levels, and sentiment analysis results, making it more interpretable and user-friendly. The

integration of visualization tools and an interactive interface ensures that users can easily understand the results and gain meaningful insights into their mental well-being.

2. METHODOLOGY

A. System Architecture

The proposed system analyzes user-generated text to detect potential mental health conditions using Natural Language Processing (NLP) and Machine Learning techniques. It consists of a user interface for input and a backend processing module for analysis. The system processes the text in multiple stages and provides a summarized result along with predictions.

B. Data Collection and Preparation

The dataset is collected from Kaggle and contains labeled text data for conditions such as depression, anxiety, and bipolar disorder. Data cleaning is performed to remove duplicates and irrelevant entries, ensuring better model performance.

C. Text Preprocessing

The input text is preprocessed by converting it to lowercase, removing special characters, links, and stop words, and normalizing the text. This improves the quality of data for further analysis.

D. Feature Extraction

TF-IDF (Term Frequency–Inverse Document Frequency) is used to convert textual data into numerical form. This helps in identifying important words that contribute to classification.

E. Model Training and Evaluation

Logistic Regression is used as the classification model due to its efficiency in text-based tasks. The model is trained using the LBFGS optimizer and evaluated using metrics like accuracy and precision to ensure reliable performance.

F. Sentiment and Behavioral Analysis

The system uses VADER sentiment analysis to detect emotional tone and combines it with machine learning predictions. It also identifies behavioral patterns such as negative thinking, stress, and social withdrawal using keyword-based rules.

G. Risk Assessment

A risk score is calculated based on sentiment, detected patterns, and classification results. The score ranges from 0 to 100, where higher values indicate higher mental health risk.

H. System Implementation

The system is developed using React.js (frontend), Node.js (backend), and Python (ML processing). It supports smooth integration between components and ensures scalability.

I. Output and User Interaction

The system displays results through a dashboard and includes a chatbot and voice input for user interaction. It also provides recommendations, helpline support, and a Safe Space Mode for user well-being.

3. TOOLS AND TECHNOLOGIES

A. Development Technologies

- **React.js:** Used for building an interactive and responsive frontend interface.
- **Node.js:** Handles backend operations and server-side logic.
- **Python:** Used for implementing machine learning and NLP components.

B. Natural Language Processing Tools

- **VADER** (Valence Aware Dictionary and Sentiment Reasoner): Used for sentiment analysis to detect emotional tone in user text.

- **TF-IDF** (Term Frequency–Inverse Document Frequency): Applied for feature extraction by converting text data into numerical form.

C. Machine Learning Models

- **Logistic Regression:** Used as the primary classification algorithm for detecting different mental health conditions.
- **DistilBERT and RoBERTa:** Transformer-based models used to enhance contextual understanding and improve classification accuracy.

D. Version Control

- **GitHub:** Used for version control, code management, and collaboration during development.

4. IMPLEMENTATION

A. Data Collection and Preprocessing

- Collected labeled dataset from Kaggle and performed data cleaning by removing duplicates and irrelevant content.
- Preprocessed text by converting to lowercase, removing special characters, URLs, and stop words to improve data quality.
- Applied tokenization and normalization to standardize the textual data for further processing.

B. Feature Extraction and Model Development

- Applied TF-IDF to transform textual data into numerical features.
- Trained a Logistic Regression model using the LBFGS optimizer for multi-class classification.
- Integrated advanced models like DistilBERT and RoBERTa to enhance contextual understanding and improve accuracy.
- Performed model tuning and validation to optimize performance.

C. Sentiment and Behavioral Analysis

- Used VADER for sentiment analysis to detect emotional tone.
- Implemented rule-based keyword detection to identify behavioral patterns such as stress, negative thinking, and social withdrawal.
- Combined sentiment scores with classification results to improve prediction reliability.

D. Risk Evaluation Mechanism

- Combined model predictions, sentiment scores, and behavioral indicators.
- Assigned weighted values to different factors for accurate assessment.
- Generated a risk score (0–100) to represent the severity of mental health conditions.

E. System Development and Integration

- Developed frontend using React.js and backend using Node.js.
- Integrated Python-based ML models via APIs to ensure smooth communication between system components.
- Implemented efficient data flow and real-time processing between modules.

F. User Interface, Testing, and Deployment

- Designed an interactive interface with chatbot and dashboard for result visualization.
- Conducted system testing to ensure accuracy, responsiveness, and reliability.
- Managed version control using GitHub and deployed the system for user access.

G. System Workflow

- The system follows a pipeline:
User Input → Preprocessing → Feature Extraction → Model Prediction → Sentiment & Behavioral Analysis → Risk Score Generation → Result Display

5. RESULTS AND FUTURE SCOPE

5.1 Result:

A. System Testing

The system was tested by integrating the frontend with the Python-based machine learning backend. User input text was processed through multiple stages including preprocessing, sentiment analysis using VADER, feature extraction using TF-IDF, classification using trained models, and risk score generation.

This end-to-end testing ensured that the system performs accurately and responds effectively to real-time user input. Additionally, various test cases were used to evaluate the system's performance across different types of text inputs, including clearly expressed and ambiguous statements.

The system demonstrated consistent performance with minimal latency, making it suitable for real-time applications. Furthermore, the integration between frontend, backend, and machine learning components was validated to ensure smooth data flow and reliable output generation.

B. Model Performance

The model was trained on a dataset of 52,678 labeled instances covering seven mental health categories. The proposed system achieved an overall accuracy of 89.97%, demonstrating strong performance in detecting mental health conditions from textual data.

The model performs well in clearly defined categories such as Normal, Anxiety, and Bipolar, while slightly lower accuracy is observed in categories like Stress and Personality Disorder due to overlapping language patterns.

Additionally, the use of transformer-based models such as DistilBERT and RoBERTa significantly improved the system's ability to understand contextual and semantic relationships within the text. The hybrid approach combining traditional machine learning and deep learning techniques enhanced overall prediction reliability.

The system also provides supporting metrics such as precision, recall, and F1-score for each category, ensuring a detailed evaluation of model performance. These results indicate that the proposed model is robust, efficient, and suitable for real-time mental health analysis applications.

C. Comparative Analysis of Models

Model	Accuracy (%)
TF-IDF + Logistic Regression	79.09
DistilBERT	83.25
RoBERTa	89.97

The comparison shows that:

- TF-IDF with Logistic Regression captures basic textual patterns
- DistilBERT improves performance using contextual understanding
- RoBERTa achieves the highest accuracy due to better language representation

D. Output and Metrics

The system provides:

- Predicted mental health category
- Confidence score :-Indicates reliability of prediction
- Risk score (0–100 scale)
- Sentiment score
- Performance metrics including **precision, recall, and F1-score** for each mental health category, indicating the accuracy and reliability of predictions

E. Real-Time Performance

The system processes user input efficiently and provides results in real time, making it suitable for practical applications such as continuous mental health monitoring. The system is designed to be

user-friendly and interpretable. Along with predictions, it provides visualizations, behavioral insights, and support suggestions, helping users better understand their mental health condition.

5.2 Future Scope:

A. Multi-Language Support — Enable the platform to analyze and respond in multiple languages to reach a broader user base.

B. Wearable Device Integration — Use data from smartwatches and fitness trackers (e.g., sleep and heart rate patterns) to enhance mental health assessment.

C. Advanced Predictive Analytics — Incorporate more sophisticated deep learning models and longitudinal analysis for improved prediction accuracy.

D. Voice and Speech Analysis: Extend the system to analyze voice inputs, tone, and speech patterns to detect emotional states more accurately.

E. Real-Time Monitoring and Alerts: Implement continuous monitoring with alert mechanisms to notify users or caregivers in high-risk situations, especially in cases of severe distress or suicidal tendencies.

6. CONCLUSION

This project presents the development of a hybrid mental health detection system that utilizes Natural Language Processing (NLP) and Machine Learning techniques to analyze user-generated text and identify potential mental health conditions at an early stage. The system integrates VADER for sentiment analysis, TF-IDF for feature extraction, and Logistic Regression along with advanced transformer models such as DistilBERT and RoBERTa to capture both contextual and semantic information effectively.

The model was trained and evaluated on a dataset of 52,678 labeled samples, achieving an overall accuracy of 89.97%. The system demonstrated strong performance in identifying well-defined categories such as anxiety, bipolar disorder, and normal mental states. However, slightly lower

performance was observed in categories like depression and suicidal tendencies due to overlapping linguistic patterns and subtle variations in expression.

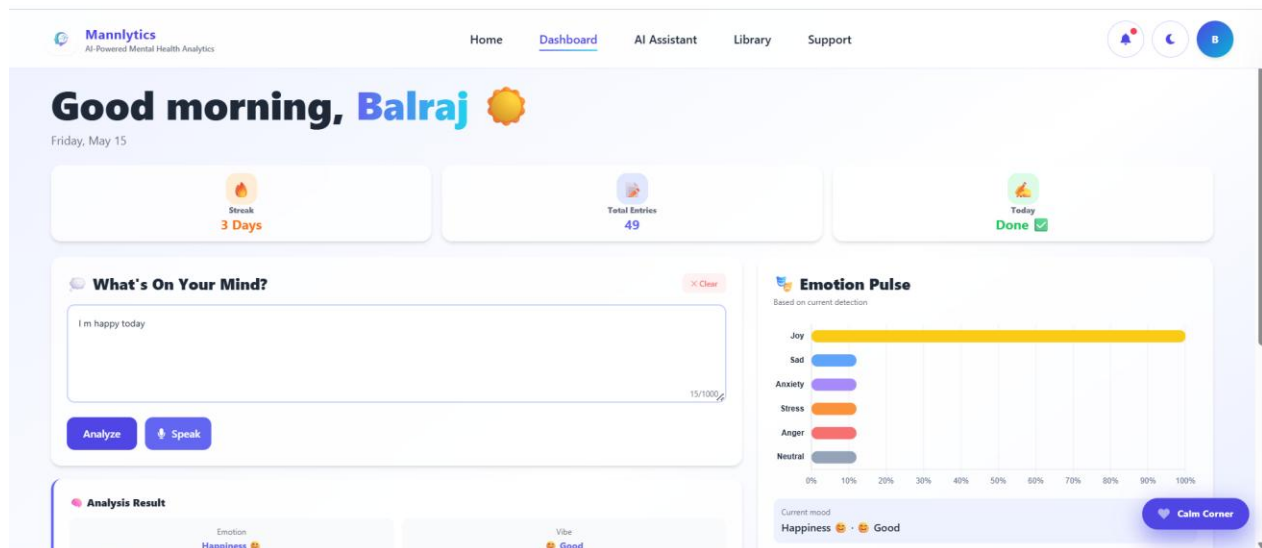
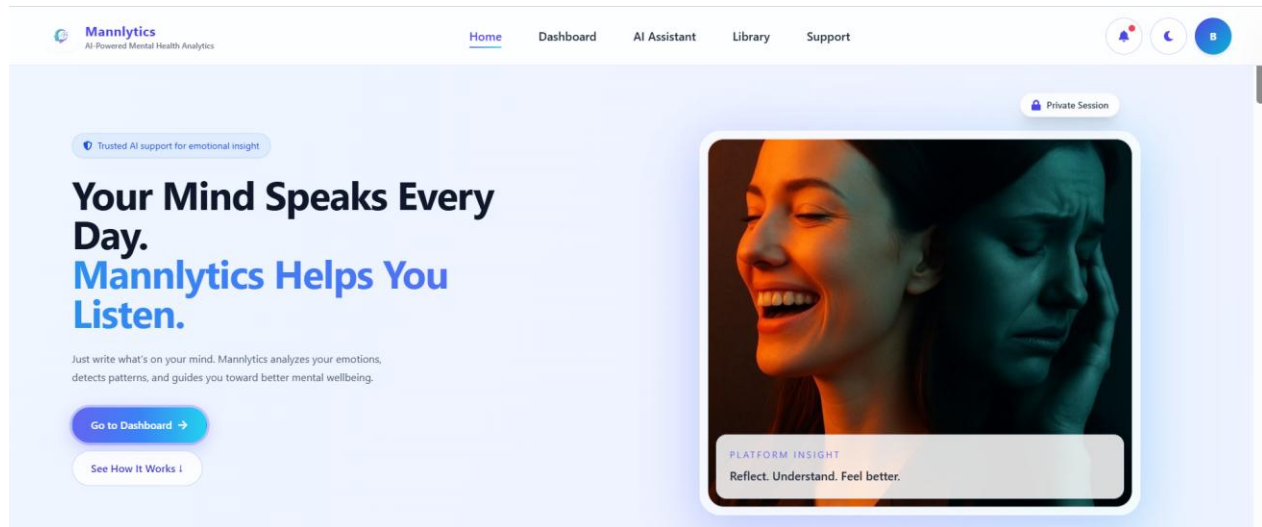
In addition to classification, the system provides meaningful insights through the generation of a risk score (scaled from 0 to 100), confidence levels, and sentiment analysis. These features enhance interpretability and help users better understand their mental state. The integration of visualization tools and a user-friendly interface further improves accessibility and usability.

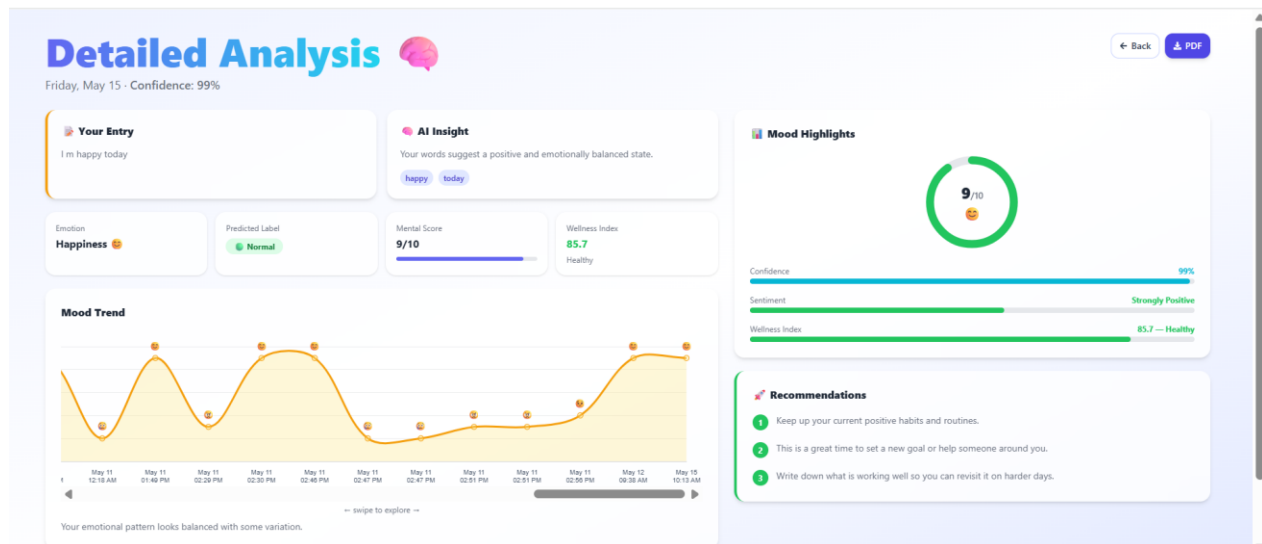
Overall, the proposed system is computationally efficient, scalable, and capable of delivering real-time analysis. It has the potential to serve as a supportive tool for early mental health assessment and awareness, contributing to timely intervention and better mental well-being.

7. REFERENCES

- [1] VADER Sentiment Analysis Documentation, *GitHub Repository*, Available: <https://github.com/cjhutto/vaderSentiment>
- [2] Scikit-learn, *Machine Learning in Python*, Available: <https://scikit-learn.org>
- [3] DistilBERT Model Documentation, Available: <https://huggingface.co/distilbert-base-uncased>
- [4] RoBERTa Model Documentation, Available: <https://huggingface.co/roberta-base>
- [5] React.js Documentation, Available: <https://react.dev>
- [6] Node.js Documentation, Available: <https://nodejs.org>
- [7] Kaggle Dataset Repository, Available: <https://www.kaggle.com>

8. APPENDICES





Mannlytics
AI-Powered Mental Health Analytics

MODES

- Mini Games
- Soothing Music
- Meditation Sessions
- Therapy Support
- Funny Videos
- Dance Videos
- Breathing Exercises
- Comfort Notes

Dark Mode

[← Back to Dashboard](#)

Search activities...

Your calm recovery zone

Slow down with soothing music, guided meditation, grounding therapy, playful mini breaks, breathing exercises, and uplifting support — all in one place.

[Start Relaxing](#)

SAFE SPACE COLLECTION

Mini Games

Shift your focus with playful mental breaks


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AI-Powered Mental Health Analytics

[Home](#)
[Dashboard](#)
[AI Assistant](#)
[Library](#)
[Support](#)





Mental Health Support & Resources

Some days feel heavier than others, and that's completely okay. Whether you need someone to listen or professional guidance — verified helplines and trusted counselors are here, ready to support you every step of the way.



Verified Mental Health Helplines

Free, confidential, and available 24x7.

Tele-MANAS
24x7 support • National service

KIRAN Mental Health Helpline
24x7 support • Multilingual

Vandrevala Foundation
24x7 support • Crisis assistance


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AI-Powered Mental Health Analytics

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[Library](#)
[Support](#)







Mood Summary

May 2026

Happiness	8x
Sadness	4x
Anger	3x
Neutral	1x
Stress	1x

17 entries this month

Last Journal

15 May 2026

"I m happy today"

Score	Confidence	Risk
0/10	99.2%	14.3

Recent Journal Entries

View All →

Fri May 15 2026	Happiness
Tue May 12 2026	Happiness
Mon May 11 2026	Stress
Mon May 11 2026	Anger
Mon May 11 2026	Anger


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 AI-Powered Mental Health Analytics

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[Support](#)





+ New Chat

RECENTS
 Today
 Hello, Feeling Low
 May 7
 Joyful Mood Boost
 May 6
 "Tiredness Feeling" "Feeling Tir...
 Managing Stress Feelings
 "Tiredness Feeling" "Feeling Tir...


Mannlytics
 Your AI Companion


Good morning, Sukriti! 🌞
 How are you feeling today?
 Share how you're feeling, pick a mood, or just start typing. I'm here to listen. ❤️

Happy

Sad

Anxious

Stressed

Tired

Strong


Mannlytics
 Hello again Sukriti! 🌞 Starting a new conversation. What's on your mind?
 11:29 AM

Share how you're feeling... (Enter to send)


Mannlytics
 AI-Powered Mental Health Analytics

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[Dashboard](#)
[AI Assistant](#)
[Library](#)
[Support](#)





+ New Chat

RECENTS
 Today
 * Happy and Content * Joyful
 Hello, Feeling Low
 May 7
 Joyful Mood Boost
 May 6
 "Tiredness Feeling" "Feeling Tir...
 Managing Stress Feelings
 "Tiredness Feeling" "Feeling Tir...


Mannlytics
 Your AI Companion


Mannlytics
 Hello Sukriti! 🌞 I'm Mannlytics, your personal mental wellness companion. How are you feeling today?
 11:29 AM

Hi there! I'm glad to hear that you're feeling happy today. Is there anything specific that's making you feel this way? Remember, it's okay to have good days and bad days, and it's important to take care of yourself. If you ever need someone to talk to, I'm here for you.

hi..i m happy today


Mannlytics
 11:30 AM

Share how you're feeling... (Enter to send)

Mannlytics

Mental Health Analysis Report

Confidence: 98%

Friday, May 15, 2026 · 11:32 AM

Prepared for: Sukriti Pahuja

Analysis Summary

Emotion: **Sadness**
Predicted Label: **Normal**
Sentiment: Strongly Negative
Wellness Index: 50.9 — Moderate

AI Insight

Your words suggest emotional heaviness or distress.

Keywords detected:

sad · today

Scores

Mental Wellness Score 2/10



Wellness Index 50.93/100



Sentiment Polarity 48/100



Confidence 98/100



Journal Entry

i m sad today

Recommendations

1. Try a relaxing activity like music, a short walk, or journaling.
2. Stay hydrated, rest well, and avoid overloading yourself today.
3. Reach out to a friend or someone you trust for a conversation.

*This report is auto-generated by Mannlytics based on your journal entry.
It is not a substitute for professional medical advice, diagnosis, or treatment.*